TokenLens: Telemetry and Spend Guard

Overview

TokenLens is the observability and cost-governance layer for Project Aegis. It is composed of two cooperating components: Telemetry and Spend Guard. Together, Telemetry and Spend Guard transform TokenLens into an end-to-end enterprise AI control plane that simultaneously observes and governs production AI usage.

Telemetry acts as the Eyes and Ears of the system. As an AI workflow runs, Telemetry silently tracks every step's resource consumption, including latency, tokens, retries, and accumulated cost, without impeding performance. It captures metadata and statistics only, and never captures customer document data.

Spend Guard acts as the Security Guard of the system. It uses the real-time cost data tracked by Telemetry to enforce policy limits deterministically. The underlying mechanism is deterministic code, not a language model, so budget enforcement is predictable and auditable. A large language model is only used to summarize context for human approvers, never to decide whether a budget has been exceeded.

Real-Time Interruption and Governance

If a runaway task or retry loop hits a budget cap, for example a policy threshold of forty rupees, Spend Guard synchronously intervenes. It interrupts the process and checkpoints its state so the task can be resumed later without losing progress.

When a response is locked because a threshold was crossed, the system requests an administrator secret key before releasing the held response to the requesting user. Every unlock attempt, whether it succeeds or fails, is written to the audit trail alongside the request identifier, the user identifier, the role of the requester, and the outcome of the attempt.

Key Question Answered: What happened? Every lock, unlock attempt, and resolution is logged with a shared request identifier, so a compliance reviewer can reconstruct the full timeline of any request after the fact.

Key Question Answered: Should this continue? Spend Guard's deterministic policy check answers this question before any further cost is incurred, rather than after the fact.

How They Work Together as an AI Control Plane

Telemetry and Spend Guard are deliberately separated. Telemetry has no authority to block or alter a request; it only observes and records. Spend Guard has no visibility into document content; it only acts on the cost and usage signals that Telemetry provides. This separation of concerns keeps each component simple, testable, and auditable on its own.

In practice, a single request generates a short, connected story: an execution log entry describing the model call, token usage, and latency; followed, only if a threshold was crossed, by one lock event and one or more unlock-attempt events, all sharing the same request identifier. Downstream dashboards join on that request identifier to reconstruct the full picture of what happened, why, and how it was resolved.

Underlying Mechanism: TokenLens captures metadata and statistics only, never customer document data. Deterministic code enforces budgets. A language model is only used to summarize context for human approvers, never to make the enforcement decision itself.

This document serves as the demo-only knowledge base for Project Aegis's retrieval-augmented terminal client, illustrating how a governed AI response should describe TokenLens, Telemetry, and Spend Guard when asked.